

Algebraic Constructions: Solution

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Assume that all of the following figures are constructed with only a straightedge and compass:

1. Is it possible to construct a square with an area equal to a given circle? **No.**

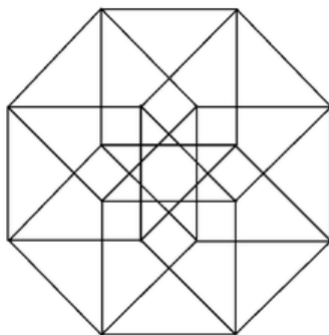
We know that π is not an algebraic number. In other words, it cannot be expressed as the root of any polynomial with integer coefficients. Given that constructible points must satisfy the roots of some polynomial (further detailed in the next solution), we know that we cannot construct a square that is some sort of multiple of π .

2. Is it possible to construct a cube double the volume of a unit cube? **No.**

It is not possible to double the volume of a unit cube. By doubling the volume of the unit cube, a new cube with volume 2 would need to be constructed. Therefore, a side with length $\sqrt[3]{2}$ would need to be constructed.

Taking a Galois approach with this, we could consider how points are formed in the first place. If we wanted to form the point $(\sqrt[3]{2}, 0)$ (or even the other pair, $(0, \sqrt[3]{2})$), then we have to consider how points in general are formed. Using a straight edge and compass, it is possible to construct circles going through two predefined points or do the same with lines, which is more useful in our case. A newly defined point can also arise from the intersection of two such lines or circles. We know from this that the points must satisfy some (at most) quadratic such that binary operations on the previous points can lead to the new point being defined. The degree of the field extension that corresponds to each point must therefore be at most 2 but at least 1. We can therefore conclude that the degree of the field extension over $\mathbb{Q}$ for any of our constructed points must be a power of 2.

However, the minimum splitting field for the polynomial $x^3 - 2$, which is the polynomial used to construct the point at $\sqrt[3]{2}$, must be $\mathbb{Q}(\sqrt[3]{2})$, which is clearly of degree 3, and this does not satisfy the premise that the constructed point must be a power of 2.



A 2D rendering of a 4D cube

3. Is it possible to construct any n -dimensional cube twice the volume of its unit representation? **Not unless n is a power of 2.**

Seeing the result we got before, we know that we can construct any point that has a degree field extension of a power of 2. This means that squares, tesseracts, and any 2^n -dimensional cube can be constructed such that they are double the area/volume of the unit cube.

Alternate reasoning is that we know that if a is constructible, so is $\sqrt{a}$. We can see that this is true because of the side lengths of a right triangle. If $\sqrt{a}$ is constructible, then so is $\sqrt[4]{a}$ and so on.

4. Is it possible to trisect any given angle using constructions? **No.**

Though there are certain angles that are obviously constructible, such as an angle of 90° , it is not possible to trisect any arbitrary one. Using a compass and straight edge, it is possible to apply any binary operations on a given angle. However, for similar reasoning as constructing lines, the field extension of a constructible angle must be of power 2.

The smallest irreducible polynomial that can be created using $\cos 20^\circ$, or an angle of 20 degrees, is $8x^3 - 6x - 1$, which is clearly of degree 3 in the rationals, implying that a 60° angle cannot be trisected. This immediately proves that any arbitrary angle cannot necessarily be trisected.